# LAB-1203: Build an Embedded AI Agent in IBM Planning Analytics using watsonx Orchestrate

Hands-on lab guide

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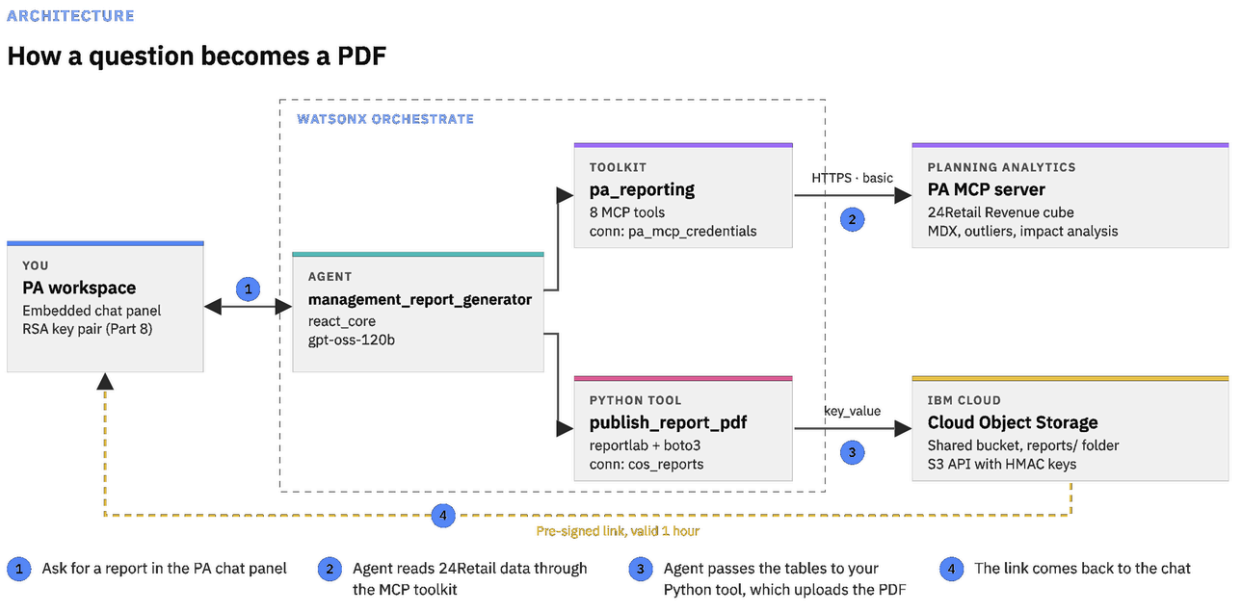
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# LAB-1203: Build a Management Report Agent on watsonx Orchestrate and embed it in Planning Analytics

## What you will build

A watsonx Orchestrate (wxO) agent that:

1. Reads live data from a Planning Analytics (PA) database through the PA MCP server.
2. Produces Budget vs Actual tables, outlier checks and variance-driver analysis.
3. Builds a PDF management report with a custom Python tool and stores it in IBM Cloud Object Storage (COS).
4. Returns a time-limited download link.
5. Runs inside the PA workspace (PAW) chat panel.




### Parts

| Part | What you do | Time |
|---|---|---|
| 0 | Prerequisites and security rules | 5 min |
| 1 | Set up the ADK and log in to wxO | 10 min |
| 2 | Connect the PA MCP server to wxO | 15 min |
| 3 | Build the agent (data and analysis only) | 15 min |
| 4 | Test the agent | 15 min |
| 5 | Create the COS connection in wxO (shared bucket) | 10 min |
| 6 | Build the report tool and add it to the agent | 20 min |
| 7 | Test the full report flow | 20 min |
| 8 | Embed the agent in the PA workspace | 25 min |
| 9 | Troubleshooting | reference |
| 10 | Clean-up | 5 min |

### Conventions in this guide

- Lines in grey code blocks are for your terminal. Where a command differs by operating system, the **Windows (PowerShell)** version comes first, then the **macOS / Linux (Terminal)** version. Use only the one for your machine. Where there is no split, the command is the same on both. Run them one block at a time.

- On Windows, open **Windows PowerShell** (Start menu, type `PowerShell`). Do not use Command Prompt (`cmd.exe`).
- `Ctrl` on Windows is `Cmd` on macOS (for example, `Ctrl+A` / `Cmd+A`).
- `<LIKE_THIS>` is a value you must replace.
- `

  *Screenshot: ...*

  ` is a placeholder for an image.
- Menu names in the cloud consoles change between releases. Where the lab author could not confirm an exact label, the step says **(confirm label)**.

---

## Getting started and environment setup

### Introduction

Welcome to LAB-1203: Build an Embedded AI Agent in IBM Planning Analytics using watsonx Orchestrate.

In this lab, you will build an AI agent that works with IBM Planning Analytics data using watsonx Orchestrate. You will configure the required connections, build and test the agent, generate a management report, and embed the agent into the Planning Analytics workspace.

Before starting the hands-on exercises, you must access the lab resources and open the required environments.

Follow the instructions in this section sequentially. The instructor will first explain the business use case and solution architecture. Begin the technical exercises only after the instructor asks you to proceed.

### Step 1: Start your workstation and open TechZone

1. You will be provided with a machine for the hands-on lab.
2. Start the machine and wait for it to finish loading.
3. Open Google Chrome.

The browser should automatically redirect you to the IBM TechZone event page.

If the event page does not open automatically:

1. Open a new Chrome tab.
2. Go to `https://techzone.ibm.com/my/events`.
3. Sign in if prompted.

### Step 2: Locate the LAB-1203 event

1. On the TechZone Events page, locate and click the **Hands-on Lab** tile.
2. Select **Tuesday** from the available event days.
3. Search for **LAB-1203**.
4. Open the event titled **Build an Embedded AI Agent in IBM Planning Analytics using watsonx Orchestrate**.
5. Review the event page and locate the lab resources and environment details.

### Step 3: Open the lab materials

The TechZone event provides the presentation and the hands-on lab guide. Locate and open the following resources.

**3.1 Instructor presentation.** Open `Lab-1203-Intro.pptx`. The instructor will use this presentation to explain the business use case, solution architecture and expected outcome.

**3.2 Hands-on lab guide.** Open `1203-Lab-Guide.pdf`. This guide contains the detailed, step-by-step instructions you will follow to complete the exercises.

**3.3 GitHub repository.** Open the LAB-1203 GitHub repository at `https://github.com/AishaDarga/LAB-1203`. Use the repository to access the supporting lab resources as directed by the instructor.

Keep the presentation and lab guide accessible throughout the session.

### Step 4: Locate your lab environment details

On the TechZone event page, scroll down to the **Attendee and Environment Management** section.

1. Locate the environment assigned to you.
2. Review the environment details and login instructions.
3. Locate the **IBM Cloud Login URL**.
4. Identify the credentials provided for accessing IBM Cloud and the lab applications.
5. Locate the access code if the event prompts you to enter one.

The access code is displayed on the screen when required. Enter it where prompted and follow the on-screen instructions.

**Important:** The event access code, IBMid, IBM Cloud password and API key are different pieces of information. Use each only for its intended purpose.

Do not share passwords, API keys or other sensitive credentials with other participants.

### Step 5: Sign in to IBM Cloud

You will use IBM Cloud to access watsonx Orchestrate.

**5.1 Open the IBM Cloud login page**

1. Return to the TechZone event page.
2. Scroll down to your assigned environment details.
3. Click **IBM Cloud Login URL**.
4. Wait for the IBM Cloud login page to open.

**5.2 Enter your login credentials**

Use the following credentials when prompted:

- **Username:** the IBMid you registered with for the lab.
- **Password:** `lab1203-pass`

Enter the credentials exactly as provided.

1. Complete the sign-in process.
2. Wait for the IBM Cloud dashboard to load.

**Important:**

- Use the IBMid associated with your lab registration.
- Enter the lab password exactly as shown: `lab1203-pass`.
- Do not use your personal IBM Cloud account unless instructed by the instructor.
- Do not share your login credentials with other participants.

If the login fails or the password is not accepted, stop and ask the instructor for assistance.

### Step 6: Launch watsonx Orchestrate

After signing in to IBM Cloud, open watsonx Orchestrate.

1. Locate the navigation menu (≡) in the upper-left corner of the IBM Cloud console.
2. Click the navigation menu.
3. Select **Resource list**.
4. Locate the **AI / Machine Learning** category.
5. Find the watsonx Orchestrate instance assigned to your lab.
6. Click the instance to open its details.
7. Click **Launch** to open watsonx Orchestrate.
8. Wait for the application to load.

Once watsonx Orchestrate opens, keep the browser tab available. You will use this environment to configure connections, register tools, build your AI agent and test its responses.

If you cannot find the watsonx Orchestrate instance or the **Launch** option, ask the instructor for assistance.

### Step 7: Access Planning Analytics (PA)

#### 7.1 Find your student ID and credentials

1. Find the student number placed on your machine desk (for example, `03`). This is your student ID.
2. On the TechZone events page, open the **lab1203-users** spreadsheet.
3. Find the row for your student ID only. Your username is `lab1203-user<ID>@labs.techzone.ibm.com`. For student 03, it is `lab1203-user03@labs.techzone.ibm.com`.
4. Note these values from your row:
   - **Username**
   - **PA Password**
   - **IBMID/VERIFY Password**

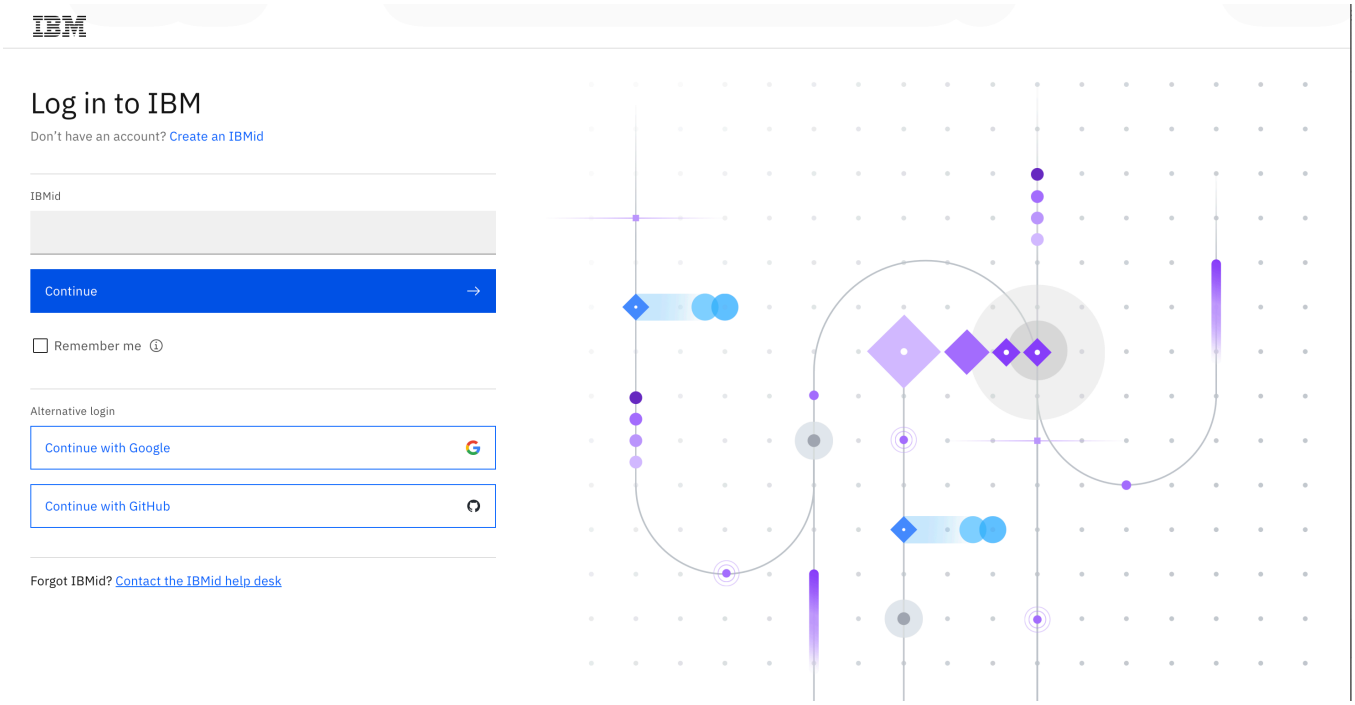
lab1203-users

| Index | Username | PA Password | IBMID/VERIFY Password | Security Verify ID | Status | Expires (Hours) | Created At |
|---|---|---|---|---|---|---|---|
| 1 | lab1203-user01@labs.techzone.ibm.com | NOT IN USE | (not provided by server) | 66300AYVAM | present | 26 | 2026-10-26T |
| 2 | lab1203-user02@labs.techzone.ibm.com | lab1203-pass | Reef-Bridge-Cabin-1376 | 66200AXNSI | created | 26 | 2026-10-26T |
| 3 | lab1203-user03@labs.techzone.ibm.com | lab1203-pass | Desert-Cedar-Storm-2305 | 66200AXNSJ | created | 26 | 2026-10-26T |
| 4 | lab1203-user04@labs.techzone.ibm.com | lab1203-pass | Moss-Eagle-Garden-3606 | 66400B1OF3 | created | 26 | 2026-10-26T |
| 5 | lab1203-user05@labs.techzone.ibm.com | lab1203-pass | Comet-Moss-Breeze-1836 | 66300AYZ1P | created | 26 | 2026-10-26T |
| 6 | lab1203-user06@labs.techzone.ibm.com | lab1203-pass | Radar-Anchor-Mountain-7496 | 66300AYZ1Q | created | 26 | 2026-10-26T |
| 7 | lab1203-user07@labs.techzone.ibm.com | lab1203-pass | Jasper-Cabin-Raven-4668 | 66400B1OF4 | created | 26 | 2026-10-26T |
| 8 | lab1203-user08@labs.techzone.ibm.com | lab1203-pass | Cedar-Star-Oak-1370 | 66400B1OF5 | created | 26 | 2026-10-26T |
| 9 | lab1203-user09@labs.techzone.ibm.com | lab1203-pass | Cedar-Shield-Robin-6911 | 66200AXNSK | created | 26 | 2026-10-26T |
| 10 | lab1203-user10@labs.techzone.ibm.com | lab1203-pass | Breeze-Spark-Island-9195 | 66300AYZ1R | created | 26 | 2026-10-26T |
| 11 | lab1203-user11@labs.techzone.ibm.com | lab1203-pass | Sierra-Cobalt-Nova-6489 | 66300AYZ1S | created | 26 | 2026-10-26T |
| 12 | lab1203-user12@labs.techzone.ibm.com | lab1203-pass | Cedar-Lagoon-Falcon-3660 | 66400B1OF6 | created | 26 | 2026-10-26T |
| 13 | lab1203-user13@labs.techzone.ibm.com | lab1203-pass | Spring-Creek-Velvet-6667 | 66300AYZ1T | created | 26 | 2026-10-26T |
| 14 | lab1203-user14@labs.techzone.ibm.com | lab1203-pass | Winter-Eagle-Eclipse-5841 | 66400B1OF7 | created | 26 | 2026-10-26T |
| 15 | lab1203-user15@labs.techzone.ibm.com | lab1203-pass | Eagle-Flint-Creek-4913 | 66400B1OF8 | created | 26 | 2026-10-26T |
| 16 | lab1203-user16@labs.techzone.ibm.com | lab1203-pass | Shield-Moss-Comet-2939 | 66100B4MTM | created | 26 | 2026-10-26T |
| 17 | lab1203-user17@labs.techzone.ibm.com | lab1203-pass | Laser-Cobalt-Anchor-9912 | 66300AYZ1U | created | 26 | 2026-10-26T |
| 18 | lab1203-user18@labs.techzone.ibm.com | lab1203-pass | Beacon-Storm-Arctic-1173 | 66200AXNSL | created | 26 | 2026-10-26T |
| 19 | lab1203-user19@labs.techzone.ibm.com | lab1203-pass | Haven-Jasper-Jade-7928 | 66100B4MTN | created | 26 | 2026-10-26T |
| 20 | lab1203-user20@labs.techzone.ibm.com | lab1203-pass | Granite-Rocket-Glacier-7927 | 66300AYZ1V | created | 26 | 2026-10-26T |
| 21 | lab1203-user21@labs.techzone.ibm.com | lab1203-pass | Orbit-Star-Prism-7329 | 66100B4MTO | created | 26 | 2026-10-26T |
| 22 | lab1203-user22@labs.techzone.ibm.com | lab1203-pass | Willow-Moon-Jade-3563 | 66100B4MTP | created | 26 | 2026-10-26T |
| 23 | lab1203-user23@labs.techzone.ibm.com | lab1203-pass | Cosmic-Swift-Timber-4177 | 66200AXNSM | created | 26 | 2026-10-26T |

Use only the row for your own student ID. Do not use another participant's credentials.

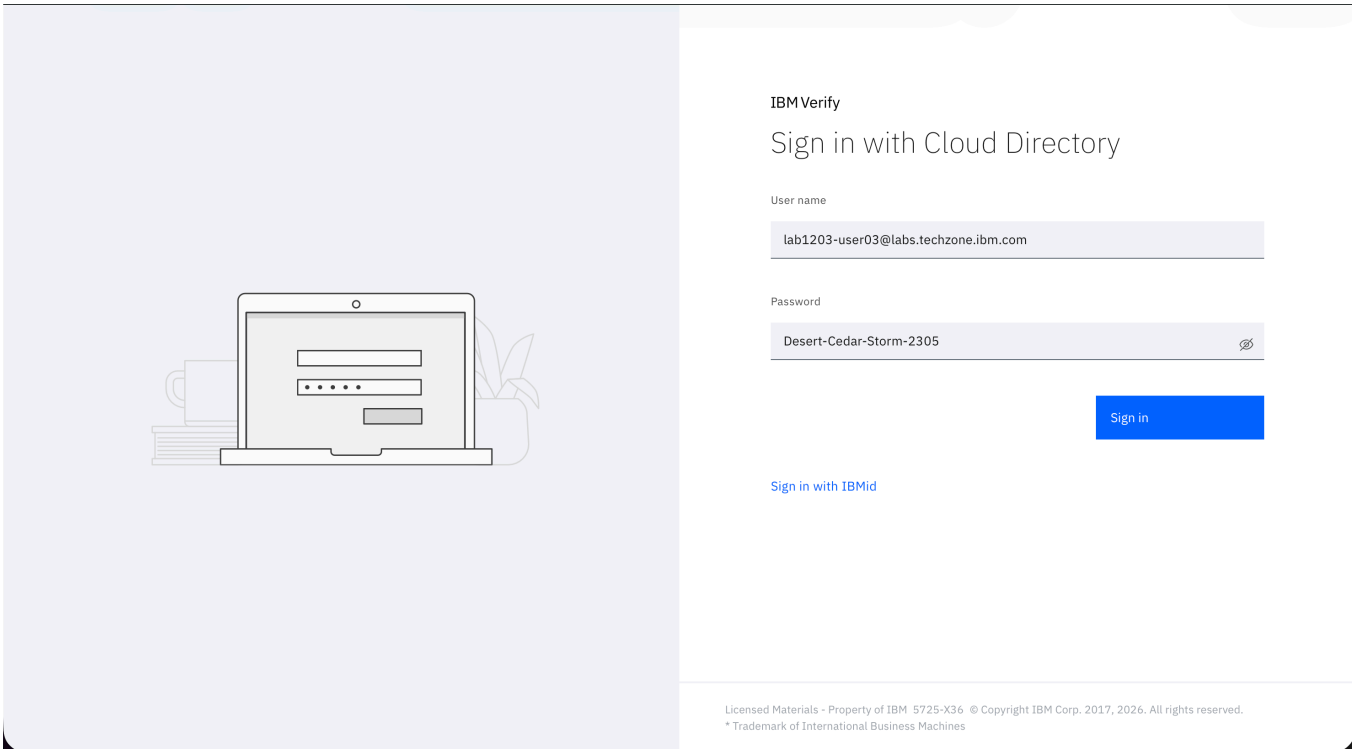
### 7.2 Sign in on the IBMid page

1. Open a new tab in Google Chrome.
2. Click the **Open IBM Planning Analytics** link provided on the TechZone event page.
3. On the **Log in to IBM** page, enter your **Username** in the **IBMid** field.
4. Click **Continue**.




### 7.3 Sign in on the IBM Verify page

1. On the **IBM Verify: Sign in with Cloud Directory** page, check that the **User name** field shows your username.
2. In the **Password** field, enter your **IBMID/VERIFY Password** from the spreadsheet.
3. Click **Sign in**.




### 7.4 Open Planning Analytics

1. If Planning Analytics asks for a password, enter your **PA Password** from the spreadsheet.
2. Wait for the Planning Analytics home page to load.

3. Confirm that you have accessed the correct lab environment.

**Important:**

- The **IBMID/VERIFY Password** (IBM Verify page) and the **PA Password** (Planning Analytics) are different values. Use each only where it is asked for.
- Enter the values exactly as shown in the spreadsheet.
- If you cannot sign in or a password is not accepted, ask the instructor for assistance.

Keep Planning Analytics and watsonx Orchestrate open in separate browser tabs, as you will use both applications during the lab.

### Step 8: Attend the instructor's use-case presentation

Before beginning the technical exercises, the instructor will explain the business use case and demonstrate the solution workflow.

During the presentation, pay attention to the following:

- The business problem the solution addresses.
- The role of IBM Planning Analytics as the source of business data.
- The role of watsonx Orchestrate in building and running the AI agent.
- How the agent analyzes financial data and identifies variance drivers.
- How the solution generates a PDF management report.
- How the agent is embedded within the Planning Analytics workspace.

The presentation provides the context you need to understand the purpose of each technical exercise.

Do not begin executing the lab guide until the instructor has completed the presentation and asks you to start.

### Lab execution sequence

| Part | Activity |
|---|---|
| Part 0 | Review prerequisites and security rules |
| Part 1 | Set up the ADK and log in to watsonx Orchestrate |
| Part 2 | Connect the Planning Analytics MCP server to watsonx Orchestrate |
| Part 3 | Build the AI agent |
| Part 4 | Test the agent's data retrieval and analysis |
| Part 5 | Create the Cloud Object Storage connection |
| Part 6 | Build the report-generation tool and add it to the agent |
| Part 7 | Test the complete management-report workflow |
| Part 8 | Embed the agent in the Planning Analytics workspace |
| Part 9 | Troubleshoot issues encountered during the lab |
| Part 10 | Clean up lab resources and credentials as instructed |

Refer to the individual sections of the lab guide for the detailed instructions for each activity.

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# Part 0: Prerequisites and security rules

### You need

| Item | Notes |
|---|---|
| IBM Cloud account with a watsonx Orchestrate instance | The lab used region `ca-tor` (Toronto) |
| Shared COS bucket details | Five values from your instructor (Part 5.1). You do not need your own COS instance |
| Planning Analytics SaaS with the sample database **24Retail** | Plus an API key that can use the PA MCP server |
| Admin access to the PA Administration area | Needed in Part 8 |
| A Windows PowerShell or a Mac / Linux terminal with Python 3.11 to 3.13 | The lab used Python 3.13 |
| `openssl` | Already installed on macOS. On Windows, install it (see "Check your tools" below) |
| The ADK ( `ibm-watsonx-orchestrate` ) | You install it yourself in Part 1.1. Nothing to do now |
| A text editor | Notepad, VS Code or any editor. Needed to create the YAML, Python and JSON files |

### Security rules for this lab

1. **Never paste API keys, HMAC keys or private keys into a chat, ticket or screenshot.** Several keys in the original build were exposed this way and had to be rotated.
2. Enter secrets in the terminal with a hidden prompt ( `Read-Host -AsSecureString` on Windows, `read -s` on macOS / Linux). The text you type is not shown and is not saved in shell history.
3. Do not store secrets in files inside the project folder. If you must create one, delete it at the end (Part 10).
4. Use credentials made for this lab and delete them afterwards.

### Check your tools (install anything missing)

Run these checks before you start Part 1. You need Python 3.11 to 3.13 and `openssl` .

**Windows (PowerShell)**

```
python --version
openssl version
```

If `python` is not found (or Windows opens the Microsoft Store), or `openssl` is not found, install them. Then **close PowerShell and open a new window** and run the checks again.

```
winget install Python.Python.3.13
winget install ShiningLight.OpenSSL.Light
```

If `winget` is not available, install Python from `https://www.python.org/downloads/` and tick **Add python.exe to PATH** in the installer. If `openssl` is still not found after reopening PowerShell, add it to the PATH for this window:

```
$env:Path += ";C:\Program Files\OpenSSL-Win64\bin"
```

**macOS / Linux (Terminal)**

```
python3 --version
openssl version
```

If `python3` is missing on macOS, install it with `brew install python@3.13`.

The ADK (`ibm-watsonx-orchestrate`) and its other Python packages are installed in Part 1.1. You do not need Docker, Node.js or the IBM Cloud CLI as a student.

*Screenshot: Lab architecture slide or the diagram above*

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## Part 1: Set up the ADK and log in to wxO

### 1.1 Create the project folder and Python environment

**Windows (PowerShell)**

```
mkdir lab-1203
cd lab-1203
python -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install ibm-watsonx-orchestrate
```

If PowerShell says that running scripts is disabled, run this once in the same window, then run the `Activate.ps1` line again:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

**macOS / Linux (Terminal)**

```
mkdir lab-1203 && cd lab-1203
python3 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install ibm-watsonx-orchestrate
```

Check it works (same on both):

```
orchestrate --version
```

You should see a version number. If you see `orchestrate : The term 'orchestrate' is not recognized` (Windows) or `command not found` (macOS), the virtual environment is not active. Activate it again with the command below.

Every time you open a new terminal, go back to the project folder and activate the environment first:

**Windows (PowerShell)**

```
cd lab-1203
.venv\Scripts\Activate.ps1
```

**macOS / Linux (Terminal)**

```
cd lab-1203
source .venv/bin/activate
```

### 1.2 Get your wxO service URL and API key

1. Sign in to IBM Cloud and open your **watsonx Orchestrate** instance.
2. Open the wxO UI. Click your profile icon, then **Settings**, then **API details**. **(confirm label)**
3. Copy the **Service instance URL**. It looks like `https://api.ca-tor.watson-orchestrate.cloud.ibm.com/instances/<INSTANCE_ID>`.
4. Click **Generate API key (confirm label)** and keep the key open. You will type it into the terminal in the next step.

*Screenshot: wxO Settings > API details page, with the URL and Generate API key button marked. Blur the key.*

### 1.3 Add and activate the environment

Add the environment. Use any name you like. The lab used `lab-1203`.

```
orchestrate env add -n lab-1203 -u "<SERVICE_INSTANCE_URL>" --type ibm_iam
```

Store the API key in a hidden variable. Run each line **on its own**. After the first line, paste the key and press Enter.

**Windows (PowerShell)**

```
$s = Read-Host "Paste your wxO API key" -AsSecureString
```

```
$WXO_KEY = [System.Net.NetworkCredential]::new("", $s).Password
```

**macOS / Linux (Terminal)**

```
read -s WXO_KEY
```

Activate (same on both):

```
orchestrate env activate lab-1203 --api-key "$WXO_KEY"
```

Remove the variable:

**Windows (PowerShell)**

```
Remove-Variable WXO_KEY, s
```

**macOS / Linux (Terminal)**

```
unset WXO_KEY
```

Check:

```
orchestrate connections list
```

You should see a table (it may be empty).

> **Token expiry.** The login lasts a limited time. If any command says `The token found for environment 'lab-1203' is missing or expired`, repeat the hidden-prompt, `activate` and remove-variable steps above.

*Screenshot: Terminal showing a successful `orchestrate env activate`*

## Part 2: Connect the PA MCP server to wxO

A **toolkit** in wxO is a bundle of tools published by an MCP server. A **connection** holds the credentials. Remote MCP servers need a **Basic** connection with these values:

| Field | Value |
|---|---|
| Username | `apiKey` (the literal word) |
| Password | your PA API key |

> **Do not use a `key_value` connection for a remote MCP toolkit.** The first attempt in this lab used `key_value` and the server answered `401 Unauthorized`.

### 2.1 Find the PA MCP server URL and API key

The PA MCP server URL has this shape:

```
https://<PA_HOST>/api/<TENANT_ID>/v0/agentic-ai/ibm-pa-tools/mcp
```

Your instructor provides `<PA_HOST>`, `<TENANT_ID>` and the PA API key.

**Instructor to confirm:** the PA screen where students create or copy the API key, and the exact URL.

*Screenshot: PA screen that shows the MCP server URL and the API key (blur the key)*

If you already have an MCP client file (for example `.bob/mcp.json`) with an `Authorization: Basic <base64>` header, the password is the part after the first colon once the base64 text is decoded. You can extract it into a hidden variable with these commands (they print nothing):

**Windows (PowerShell)**

```
$h = (Get-Content .bob\mcp.json -Raw | ConvertFrom-Json).mcpServers.'ibm-pa-
tools'.headers.Authorization.Split(" ",2)[1]
$PA_KEY = [Text.Encoding]::UTF8.GetString([Convert]::FromBase64String($h)).Split(":",2)[1]
Remove-Variable h
```

**macOS / Linux (Terminal)**

```
PA_KEY=$(python3 -c 'import json,base64;d=json.load(open(".bob/mcp.json"))["mcpServers"]["ibm-
pa-tools"]["headers"]["Authorization"].split(" ",1)
[1];print(base64.b64decode(d).decode().split(":",1)[1])')
```

Otherwise, type it in by hand. Run each line **on its own**:

**Windows (PowerShell)**

```
$s = Read-Host "Paste your PA API key" -AsSecureString
```

```
$PA_KEY = [System.Net.NetworkCredential]::new("", $s).Password
```

**macOS / Linux (Terminal)**

```
read -s PA_KEY
```

### 2.2 Create the connection

Create the connection record:

```
orchestrate connections add -a pa_mcp_credentials
```

Configure it for both environments. The `--server-url` is the PA MCP URL:

**Windows (PowerShell)**

```
foreach ($e in "draft","live") { orchestrate connections configure -a pa_mcp_credentials --env
$e --type team --kind basic --server-url "<PA_MCP_URL>" }
```

**macOS / Linux (Terminal)**

```
for env in draft live; do orchestrate connections configure -a pa_mcp_credentials --env $env --
type team --kind basic --server-url "<PA_MCP_URL>"; done
```

Set the credentials:

**Windows (PowerShell)**

```
foreach ($e in "draft","live") { orchestrate connections set-credentials -a pa_mcp_credentials
--env $e -u apiKey -p "$PA_KEY" }
```

**macOS / Linux (Terminal)**

```
for env in draft live; do orchestrate connections set-credentials -a pa_mcp_credentials --env
$env -u apiKey -p "$PA_KEY"; done
```

Clear the variable:

**Windows (PowerShell)**

```
Remove-Variable PA_KEY, s -ErrorAction SilentlyContinue
```

**macOS / Linux (Terminal)**

```
unset PA_KEY
```

Check:

```
orchestrate connections list
```

`pa_mcp_credentials` should show as configured for draft and live.

### 2.3 Register the toolkit with eight scoped tools

The PA server offers about 50 tools. A small agent works better with a short, focused list, so import only the tools it needs.

```
orchestrate toolkits add --kind mcp --name pa_reporting --description "Planning Analytics read-
only reporting tools" --url "<PA_MCP_URL>" --transport streamable_http --tools
"execute_mdx_and_get_view,get_cube_sample_members,lookup_potential_members,perform_outlier_dete
ction,get_outlier_summary,perform_impact_analysis,get_impact_analysis_summary,generate_explorat
ion_analysis_report" --app-id pa_mcp_credentials
```

Check:

**Windows (PowerShell)**

```
orchestrate toolkits list
orchestrate tools list | Select-String pa_reporting
```

**macOS / Linux (Terminal)**

```
orchestrate toolkits list
orchestrate tools list | grep pa_reporting
```

Tool names have the form `pa_reporting:<tool>`.

> `generate_exploration_analysis_report` is imported here but the final agent does **not** use it. During the build it returned `HTTP 500` from the PA server regardless of the input. Part 6 replaces it with a custom tool.

### 2.4 Look at the data (optional but useful)

In wxO, open **Manage agents** or the **Tools** catalog **(confirm label)** and find `pa_reporting`. Open `execute_mdx_and_get_view` and read its description.

*Screenshot: wxO tool catalog showing the pa_reporting tools*

**About the 24Retail Revenue cube**

| Dimension | Members used in this lab |
|---|---|
| organization | Total Company, East Region (100), Central Region (200), West Region (300), Canada (400) |
| Channel | Channel Total, Retail (10), Internet (20), Distribution (30) |
| product | Product Total, Phones (20000), PCs (30000), Tablets (40000) |
| Month | Year (full year), Q1 to Q4, Jan to Dec |
| Year | Y1 = 2022, Y2 = 2023, Y3 = 2024 |
| Version | Actual, Version 1 (Budget), Forecast, Target, Prior Year Actual |
| Revenue (measures) | Gross Revenue, Cost of Sales, Gross Margin, Gross Margin %, Units Sold |

The cube does not state its currency, so the agent must not print `$`.

---

## Part 3: Build the agent (data and analysis only)

In this part the agent can read data, find outliers and explain variance. PDF output is added in Part 6.

### 3.1 Create the file

**Windows (PowerShell)**

```
New-Item -ItemType Directory -Force agents, tools
```

**macOS / Linux (Terminal)**

```
mkdir -p agents tools
```

Create `agents/management_report_generator.agent.yaml` with this exact content. Use any text editor (Notepad, VS Code). On Windows, check that the file name does not end in `.txt` (in Notepad choose **Save as type: All files**) and that the encoding is **UTF-8**. The same applies to every file you create in this lab.

Shortcut: the finished files are also in the LAB-1203 GitHub repository (`https://github.com/AishaDarga/LAB-1203`). You may download the ZIP (**Code**, then **Download ZIP**) and copy the `agents` and `tools` folders into `lab-1203` instead of typing them. If you have Git installed, `git clone https://github.com/AishaDarga/LAB-1203` also works.

```yaml
spec_version: v1
kind: native
name: management_report_generator
description: >
  Generates Budget vs Actual management reports for the 24Retail Planning
  Analytics database. Pulls revenue and margin data from the Revenue cube, flags
  anomalies and explains variance drivers.
llm: groq/openai/gpt-oss-120b
style: react_core
instructions: |
  You are the 24Retail Management Report Generator, an agent operating within watsonx
Orchestrate.
  You produce finance management reports from the Planning Analytics server "24Retail".

  Behavior and sources:
  - Always use the available tools to retrieve data before relying on your own knowledge.
  - Never invent numbers. If a tool does not return a value, say so.
  - If period, version or entity is missing, ask for all missing items in a single question.
    Defaults when the user does not say: entity Total Company, Channel Total, Product Total,
version Actual compared with Version 1 (Budget).
  - Label all narrative commentary "Draft - for review".

  Data rules for the Revenue cube on server 24Retail:
  - Dimensions: organization, Channel, product, Month, Year, Version, Revenue.
  - Year members: Y1 = 2022, Y2 = 2023, Y3 = 2024.
  - Version members include Actual, Version 1 (Budget), Forecast, Target.
  - Revenue measures include Units Sold, Gross Revenue, Cost of Sales, Gross Margin, Gross
Margin %.
  - Month member "Year" is the full-year total.
  - Use pa_reporting:lookup_potential_members or pa_reporting:get_cube_sample_members when
unsure of a member name.

  Tool usage rules:
  - Use pa_reporting:execute_mdx_and_get_view to read data. Keep result sets small (summary
level first, drill down only when asked).
  - Use pa_reporting:perform_outlier_detection and pa_reporting:get_outlier_summary on the view
state returned by execute_mdx_and_get_view to find unusual values. Always report outliers in an
Exceptions section.
  - Use pa_reporting:perform_impact_analysis and pa_reporting:get_impact_analysis_summary to
explain what drives a variance.
  - Before building any breakdown table (region, channel, product), verify that the rows add up
to the Total Company figure within 0.1 M. If they do not, re-query the mismatched member using
its cube member ID. Region member IDs are: East Region=100, Central Region=200, West
Region=300, Canada=400. Always use member IDs, not aliases, in MDX.
  - Call tools with all available information. If a tool returns an error, read the message and
retry with corrections. After 2 failed attempts, tell the user and suggest alternatives.

  Report structure (for full reports):
  1. Executive summary: headline Gross Revenue, Gross Margin %, variance versus Budget.
  2. Actual versus Budget, Forecast and Target table with variance and variance percent.
  3. Margin analysis: units sold, cost of sales, gross margin.
  4. Exceptions: outliers and data quality warnings.
  5. Draft commentary on the variance drivers.

  Formatting:
  - Always show monetary and unit amounts in millions (M) with one decimal, for every row in
every table without exception.
  - Do not assume a currency symbol. The cube does not state its currency, so omit "$" unless
the user supplies one.
  - Whenever two versions or years are compared, always include Variance and Variance %
columns.
  - Calculate Variance and Variance % from the full-precision values returned by the tool, then
round to one decimal only for display. Never subtract numbers that are already rounded.
```

```
  - When reporting outliers, list each flagged value once, make the count match the rows
listed, and keep every table row to the same number of columns.
  - Variance (M) = Version value minus Budget value. Variance % = (Variance / Budget) × 100.
Compute these from the full-precision numbers returned by the tools, then round to one decimal
for display. These two columns may be computed; all other columns must come directly from tool
results.
  - Never label a value as "derived". Retrieve all measures directly from tool results. If a
tool did not return a value, call the tool again or state the value is unavailable.
  - Keep chat replies concise. A full report may be longer.
tools:
  - pa_reporting:execute_mdx_and_get_view
  - pa_reporting:get_cube_sample_members
  - pa_reporting:lookup_potential_members
  - pa_reporting:perform_outlier_detection
  - pa_reporting:get_outlier_summary
  - pa_reporting:perform_impact_analysis
  - pa_reporting:get_impact_analysis_summary
collaborators: []
```

**Why these choices**

| Setting | Reason |
|---|---|
| `llm: groq/openai/gpt-oss-120b` | The model used in the lab |
| `style: react_core` | This model forces `react_core`. The `planner` style does not apply to it |
| 7 tools, no collaborators | A small tool list keeps the model accurate |
| Region IDs in the instructions | Without them the agent once looked up "Canada" by alias and returned the wrong number |
| Variance rule | Two earlier rules contradicted each other ("never compute figures" and "always show variance"). The model then showed `N/A` |

### 3.2 Import the agent

```
orchestrate agents import -f agents/management_report_generator.agent.yaml
```

Expected output ends with `Agent 'management_report_generator' created successfully`.

### 3.3 Find it in the wxO UI

1. Open the wxO UI.
2. Open the left menu and choose **Build** then **Agent Builder (confirm label)**.
3. Select **management_report_generator**.
4. On the agent page, check the **Toolset** section lists the seven `pa_reporting` tools.

*Screenshot: Agent builder page for management_report_generator with the toolset visible*

---

## Part 4: Test the agent (data only)

> Test in the wxO browser **preview** panel. The command `orchestrate chat ask` often times out after 90 to 180 seconds on the first call, so avoid it.

### 4.1 Open the preview

On the agent page, use the **Preview** panel on the right **(confirm label)**.

*Screenshot: Preview panel with an empty chat*

### 4.2 Run these prompts, one message at a time, in the same conversation

**Prompt 1: executive summary**

```
Show me Total Company full-year 2024 Gross Revenue, Gross Margin and Gross Margin % for Actual, Budget (Version 1) and Prior Year Actual. Amounts in millions with one decimal. Include Variance vs Budget (M) and Variance % vs Budget columns.
```

**Prompt 2: regional breakdown**

```
Break down 2024 full-year Actual and Budget Gross Revenue by region using member IDs 100 (East Region), 200 (Central Region), 300 (West Region), 400 (Canada). Millions, with Variance vs Budget and Variance %. Check that the four regions sum to Total Company 76.1 M.
```

**Prompt 3: outliers**

```
Run outlier detection on the 2024 Total Company full-year data across Actual, Budget, Prior Year Actual, Forecast and Target. Summarise what is flagged.
```

**Prompt 4: variance drivers**

```
Run impact analysis on Actual vs Budget for 2024 and summarise the top variance drivers.
```

### 4.3 Check the answers against the cube

These values were checked directly in the PA cube (Gross Revenue, full year 2024):

| Member | Actual | Budget (Version 1) |
|---|---|---|
| Total Company | 76,101,225 | 88,464,390 |
| East Region (100) | 20,100,102 | 25,228,094 |
| Central Region (200) | 13,998,373 | 16,068,012 |
| West Region (300) | 19,131,048 | 21,659,038 |
| Canada (400) | 22,871,703 | 25,509,245 |
| Total Company, Prior Year Actual | 73,606,160 | |

In millions with one decimal, the regions must add up to Total Company: 20.1 + 14.0 + 19.1 + 22.9 = 76.1 for Actual and 25.2 + 16.1 + 21.7 + 25.5 = 88.5 for Budget.

**Known data quirks (not agent errors)**

- Target 2023 is about 713.5 M and Forecast 2022 is about 217.9 M. These are far from the other years and are the outliers the detector should flag if you include those years.
- Variance may differ by 0.1 from a hand calculation on rounded numbers. The agent is told to compute from full-precision values.

*Screenshot: Preview showing the Actual vs Budget table*

### 4.4 Optional: see the tool calls

In the preview, expand the steps (shown as "Step 1, Step 2..." under the answer) to see which tools were called and with what MDX **(confirm label)**.

---

## Part 5: Create the COS connection in wxO

The PA server has a built-in PDF tool but it failed with `HTTP 500` during the build. So you will build your own tool that:

1. creates the PDF,
2. uploads it to Cloud Object Storage (COS),
3. returns a link.

All students use **one shared COS bucket** that the instructor created. You do not create a bucket or a credential. You only store the instructor's values in a wxO connection.

### 5.1 Get the five values from your instructor

| Key | Value |
|---|---|
| `COS_ENDPOINT` | given by the instructor |
| `COS_REGION` | given by the instructor (for example `ca-tor`) |
| `COS_BUCKET` | given by the instructor |
| `COS_ACCESS_KEY_ID` | given by the instructor |
| `COS_SECRET_ACCESS_KEY` | given by the instructor |

Treat the access key and secret like a password. Do not paste them into chat, tickets or screenshots.

Because the bucket is shared, every student's reports go to the same `reports/` folder. File names contain a timestamp and a random id, so they do not overwrite each other. The agent gives you a link to your own file, valid for 1 hour.

### 5.2 Create the wxO connection

The Python tool reads a `key_value` connection named `cos_reports` with the five entries from 5.1.

Create and configure it:

```
orchestrate connections add -a cos_reports
orchestrate connections configure -a cos_reports --env draft --type team --kind key_value
orchestrate connections configure -a cos_reports --env live --type team --kind key_value
```

Enter the two secrets without displaying them. Run each line **on its own**:

**Windows (PowerShell)**

```
$s = Read-Host "COS access key id" -AsSecureString
```

```
$COS_AK = [System.Net.NetworkCredential]::new("", $s).Password
```

```
$s = Read-Host "COS secret access key" -AsSecureString
```

```
$COS_SK = [System.Net.NetworkCredential]::new("", $s).Password
```

**macOS / Linux (Terminal)**

```
read -s COS_AK
```

```
read -s COS_SK
```

Set the credentials:

**Windows (PowerShell)**

```
foreach ($e in "draft","live") { orchestrate connections set-credentials -a cos_reports --env
$e -e "COS_ENDPOINT=<ENDPOINT>" -e "COS_REGION=<REGION>" -e "COS_BUCKET=<BUCKET>" -e
"COS_ACCESS_KEY_ID=$COS_AK" -e "COS_SECRET_ACCESS_KEY=$COS_SK" }
```

**macOS / Linux (Terminal)**

```
for e in draft live; do orchestrate connections set-credentials -a cos_reports --env $e -e
"COS_ENDPOINT=<ENDPOINT>" -e "COS_REGION=<REGION>" -e "COS_BUCKET=<BUCKET>" -e
"COS_ACCESS_KEY_ID=$COS_AK" -e "COS_SECRET_ACCESS_KEY=$COS_SK"; done
```

Clear the variables:

**Windows (PowerShell)**

```
Remove-Variable COS_AK, COS_SK, s
```

**macOS / Linux (Terminal)**

```
unset COS_AK COS_SK
```

Check:

```
orchestrate connections list
```

`cos_reports` should show `key_value` with credentials set for draft and live.

---

## Part 6: Build the report tool and add it to the agent

### 6.1 Create `tools/requirements.txt`

```
reportlab
boto3
```

### 6.2 Create `tools/publish_report_pdf.py`

```
import io
import uuid
from datetime import datetime, timezone
from typing import List

from pydantic import BaseModel, Field
from ibm_watsonx_orchestrate.agent_builder.tools import tool
from ibm_watsonx_orchestrate.agent_builder.connections import ConnectionType
from ibm_watsonx_orchestrate.run import connections

COS_APP_ID = "cos_reports"
LINK_TTL_SECONDS = 3600


class ReportTable(BaseModel):
    heading: str = Field(description="Section heading shown above the table.")
```

```
    columns: List[str] = Field(description="Column headers.")
    rows: List[List[str]] = Field(
        description="Table rows. Every row must have the same number of cells as columns. All
values are strings, already formatted (for example '92.6')."
    )
    summary: str = Field(
        default="",
        description=(
            "Optional 2-4 sentence explanation shown above the table. "
            "State what the table shows, call out the biggest variance and explain why it
matters. "
            "Do not repeat numbers already in the table rows."
        ),
    )


_CHAR_MAP = {
    "\u2010": "-", "\u2011": "-", "\u2012": "-", "\u2212": "-",
    "\u00a0": " ", "\u202f": " ", "\u2009": " ", "\u200b": "",
}


def _safe(text: str) -> str:
    """Replace characters the built-in Helvetica font cannot draw (they render as black
boxes)."""
    for src, dst in _CHAR_MAP.items():
        text = text.replace(src, dst)
    return text.encode("cp1252", errors="replace").decode("cp1252")


def _inline(text: str) -> str:
    """Escape XML, then turn **bold** and *italic* markdown into reportlab tags."""
    import re
    from xml.sax.saxutils import escape

    out = escape(_safe(text))
    out = re.sub(r"\*\*(.+?)\*\*", r"<b>\1</b>", out)
    out = re.sub(r"(?<!\*)\*(?!\s)(.+?)(?<!\s)\*(?!\*)", r"<i>\1</i>", out)
    return out


def _build_pdf(title: str, subtitle: str, tables: List[ReportTable], commentary: str) ->
bytes:
    import re
    from reportlab.lib import colors
    from reportlab.lib.pagesizes import A4
    from reportlab.lib.styles import getSampleStyleSheet
    from reportlab.lib.units import mm
    from reportlab.platypus import Paragraph, SimpleDocTemplate, Spacer, Table, TableStyle

    buffer = io.BytesIO()
    doc = SimpleDocTemplate(
        buffer, pagesize=A4, leftMargin=18 * mm, rightMargin=18 * mm,
        topMargin=18 * mm, bottomMargin=18 * mm, title=title,
    )
    styles = getSampleStyleSheet()
    story = [Paragraph(_inline(title), styles["Title"])]
    if subtitle:
        story.append(Paragraph(_inline(subtitle), styles["Normal"]))
    story.append(Spacer(1, 6 * mm))

    col_width = (A4[0] - 36 * mm) / 6

    for t in tables:
```

```
        story.append(Paragraph(_inline(t.heading), styles["Heading2"]))
        if t.summary:
            story.append(Paragraph(_inline(t.summary), styles["Normal"]))
            story.append(Spacer(1, 3 * mm))
        data = [[_safe(c) for c in t.columns]] + [[_safe(c) for c in r] for r in t.rows]
        n_cols = len(t.columns)
        col_widths = [col_width * 2] + [col_width] * (n_cols - 1) if n_cols > 1 else None
        table = Table(data, repeatRows=1, colWidths=col_widths)
        table.setStyle(TableStyle([
            ("BACKGROUND", (0, 0), (-1, 0), colors.HexColor("#161616")),
            ("TEXTCOLOR", (0, 0), (-1, 0), colors.white),
            ("FONTNAME", (0, 0), (-1, 0), "Helvetica-Bold"),
            ("FONTSIZE", (0, 0), (-1, -1), 9),
            ("ALIGN", (1, 0), (-1, -1), "RIGHT"),
            ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
            ("GRID", (0, 0), (-1, -1), 0.25, colors.HexColor("#c6c6c6")),
            ("ROWBACKGROUNDS", (0, 1), (-1, -1), [colors.white, colors.HexColor("#f4f4f4")]),
            ("TOPPADDING", (0, 0), (-1, -1), 4),
            ("BOTTOMPADDING", (0, 0), (-1, -1), 4),
        ]))
        story.append(table)
        story.append(Spacer(1, 8 * mm))

    if commentary:
        story.append(Paragraph("Executive Commentary - Draft - for review",
styles["Heading2"]))
        for para in commentary.split("\n"):
            line = para.strip()
            bare = re.sub(r"[\s*_.\-:]+", " ", _safe(line)).strip().lower()
            if not line or bare in ("draft for review", "executive commentary draft for
review"):
                continue
            if line[:2] in ("- ", "* "):
                line = "\u2022 " + line[2:]
            story.append(Paragraph(_inline(line).replace("\u2022", "&bull;"),
styles["Normal"]))
            story.append(Spacer(1, 2 * mm))

    generated = datetime.now(timezone.utc).strftime("%Y-%m-%d %H:%M UTC")

    def footer(canvas, _doc):
        canvas.saveState()
        canvas.setFont("Helvetica", 8)
        canvas.drawString(18 * mm, 10 * mm, f"Generated {generated} - AI-assisted - Draft -
for review")
        canvas.drawRightString(A4[0] - 18 * mm, 10 * mm, f"Page {_doc.page}")
        canvas.restoreState()

    doc.build(story, onFirstPage=footer, onLaterPages=footer)
    return buffer.getvalue()


@tool(
    name="publish_report_pdf",
    description=(
        "Builds a PDF management report from already-retrieved table data and commentary,
uploads it to "
        "IBM Cloud Object Storage and returns a time-limited download link. Use only after the
data has been "
        "retrieved from Planning Analytics."
    ),
    expected_credentials=[{"app_id": COS_APP_ID, "type": ConnectionType.KEY_VALUE}],
)
def publish_report_pdf(
```

```
    title: str,
    tables: List[ReportTable],
    commentary: str = "",
    subtitle: str = "",
) -> dict:
    """Create a PDF report, store it in Cloud Object Storage and return a download link.

    Args:
        title (str): Report title, for example '24Retail FY2024 Management Report'.
        tables (List[ReportTable]): Tables to include. Each table has a heading, columns, rows
and an optional summary. The summary is a 2-4 sentence explanation shown above the table: state
what the table shows, call out the biggest variance and explain its significance. Do not repeat
numbers already in the rows.
        commentary (str): Full executive commentary for the end of the report. Write 1-2
paragraphs covering overall performance, key risks and outlook. Separate paragraphs with a
newline. Do not include the literal phrase 'Draft - for review' — the tool adds that label
automatically.
        subtitle (str): One line under the title, for example the entity, version and period.

    Returns:
        dict: download_url (valid for one hour), file_name and expires_in_minutes.
    """
    import boto3
    from botocore.config import Config

    # Normalise: the runtime may deliver each table as a plain dict instead of a ReportTable
object.
    normalised = []
    for t in tables:
        if isinstance(t, dict):
            normalised.append(ReportTable(**t))
        else:
            normalised.append(t)
    tables = normalised

    for t in tables:
        for r in t.rows:
            if len(r) != len(t.columns):
                raise ValueError(
                    f"Table '{t.heading}' has a row with {len(r)} cells but {len(t.columns)}
columns."
                )

    creds = connections.key_value(COS_APP_ID)
    client = boto3.client(
        "s3",
        endpoint_url=creds["COS_ENDPOINT"],
        aws_access_key_id=creds["COS_ACCESS_KEY_ID"],
        aws_secret_access_key=creds["COS_SECRET_ACCESS_KEY"],
        config=Config(signature_version="s3v4", s3={"addressing_style": "path"}),
        region_name=creds.get("COS_REGION", "us-south"),
    )
    bucket = creds["COS_BUCKET"]

    pdf_bytes = _build_pdf(title, subtitle, tables, commentary)
    stamp = datetime.now(timezone.utc).strftime("%Y%m%d-%H%M%S")
    key = f"reports/{stamp}-{uuid.uuid4().hex[:8]}.pdf"

    client.put_object(Bucket=bucket, Key=key, Body=pdf_bytes, ContentType="application/pdf")
    url = client.generate_presigned_url(
        "get_object",
        Params={
            "Bucket": bucket,
            "Key": key,
```

```
            "ResponseContentType": "application/pdf",
            "ResponseContentDisposition": f'inline; filename="{key.split("/")[-1]}"',
        },
        ExpiresIn=LINK_TTL_SECONDS,
    )
    return {"download_url": url, "file_name": key, "expires_in_minutes": LINK_TTL_SECONDS //
60}
```

**How the tool works**

| Piece | What it does |
|---|---|
| `@tool(... expected_credentials=...)` | Declares that the tool needs the `cos_reports` key_value connection |
| `ReportTable` | The shape of each table the agent passes in: heading, columns, rows, summary |
| dict-to-`ReportTable` loop | The wxO runtime passes tables as plain dictionaries, which caused `AttributeError: 'dict' object has no attribute 'rows'` until this loop was added |
| `_safe` | Replaces characters Helvetica cannot draw. A non-breaking hyphen printed as a black box before this |
| `_inline` | Turns `**bold**` and `*italic*` markers into real formatting |
| `generate_presigned_url` | Makes a private link that expires after 3600 seconds |

### 6.3 (Optional) Test the PDF code on your computer

This checks the layout without using wxO or COS:

**Windows (PowerShell)**

```
pip install reportlab
cd tools
python -c "import publish_report_pdf as m; T=m.ReportTable;
open('test.pdf','wb').write(m._build_pdf('Test','Total Company',
[T(heading='Regions',summary='East is lowest.',columns=['Region','Actual (M)'],rows=
[['East','20.1']])],'Overall revenue missed budget.'))"
Start-Process test.pdf
cd ..
```

**macOS / Linux (Terminal)**

```
pip install reportlab
cd tools
python3 -c "import publish_report_pdf as m; T=m.ReportTable;
open('/tmp/test.pdf','wb').write(m._build_pdf('Test','Total Company',
[T(heading='Regions',summary='East is lowest.',columns=['Region','Actual (M)'],rows=
[['East','20.1']])],'Overall revenue missed budget.'))"
cd ..
open /tmp/test.pdf
```

### 6.4 Import the tool

```
orchestrate tools import -k python -f tools/publish_report_pdf.py -r tools/requirements.txt --
app-id cos_reports
```

Expected output ends with `Tool 'publish_report_pdf' imported successfully`. If you import it again later, it says `Updating...`.

Check:

**Windows (PowerShell)**

```
orchestrate tools list | Select-String publish_report_pdf
```

**macOS / Linux (Terminal)**

```
orchestrate tools list | grep publish_report_pdf
```

### 6.5 Add the tool to the agent: three edits to the YAML

Open `agents/management_report_generator.agent.yaml` and make these three changes.

**Edit 1.** Replace the description:

```
description: >
  Generates Budget vs Actual management report packs for the 24Retail Planning
  Analytics database. Pulls revenue and margin data from the Revenue cube, flags
  anomalies, explains variance drivers and publishes a PDF to Cloud Object Storage.
```

**Edit 2.** In `instructions`, directly after the line that begins `- Before building any breakdown table`, insert this bullet. It must be indented by two spaces like the lines around it, and it is one long line:

```
  - When the user asks for a PDF or report, call publish_report_pdf. Populate tables from data
already retrieved in this conversation. If any data is missing, call
pa_reporting:execute_mdx_and_get_view to fetch it before building the PDF. Pass all tables with
heading, columns, rows and a summary. Every row must have exactly the same number of cells as
columns. All values must be pre-formatted strings in millions with one decimal (for example
"92.6"), no currency symbol. Include Variance and Variance % columns whenever two versions or
periods are compared. The summary field for each table must be 2-4 sentences: state what the
table shows, name the member with the largest unfavourable variance and explain its business
significance, and name the member with the best performance if applicable. Set commentary to a
2-3 paragraph executive narrative covering overall performance versus Budget and Prior Year,
the main variance drivers, any data-quality caveats, and suggested next steps for management.
Use only figures already retrieved. Do not include the literal phrase "Draft - for review" in
the commentary or any summary string; the tool adds that label automatically. Set subtitle to
the entity, version and period. The tool returns download_url, file_name and
expires_in_minutes. Present the download_url as a clickable link with the text "Download PDF
(valid for 1 hour)". Never re-generate the link unless the user asks again.
```

**Edit 3.** At the end of the `tools:` list, add:

```
  - publish_report_pdf
```

Re-import:

```
orchestrate agents import -f agents/management_report_generator.agent.yaml
```

Expected output: `Existing Agent 'management_report_generator' found. Updating...` and `updated successfully`.

*Screenshot: Agent builder toolset showing publish_report_pdf added*

---

## Part 7: Test the full report flow

Use the preview panel. Start a **new conversation** (reset the chat) so old context does not interfere.

### 7.1 Run the seven prompts in order, in one conversation

1. Executive summary, 2. Regional, 3. Channel and 4. Product use the prompts from Part 4. Add these:

**Prompt 3: channel breakdown**

```
Break down 2024 full-year Actual and Budget Gross Revenue by channel: Retail (10), Internet (20), Distribution (30). Millions, Variance vs Budget and Variance %.
```

**Prompt 4: product category**

```
Break down 2024 full-year Actual and Budget Gross Revenue by product category: Phones (20000), PCs (30000), Tablets (40000). Millions, Variance vs Budget and Variance %.
```

**Prompt 5: quarterly trend**

```
Show 2024 quarterly Actual and Budget Gross Revenue for Q1, Q2, Q3, Q4 at Total Company level. Millions.
```

**Prompt 6: exceptions and drivers**

```
Run outlier detection on the 2024 Total Company full-year data across Actual, Budget, Prior Year Actual, Forecast and Target. Then run impact analysis on Actual vs Budget and summarise the top variance drivers.
```

**Prompt 7: publish the PDF**

```
Generate a PDF report titled "24Retail FY2024 Management Report", subtitle "Total Company - Full Year - All Channels - All Products". Include all tables from this conversation. For each table write a 3-sentence summary explaining what it shows, the biggest unfavourable variance and its business significance. Write a 3-paragraph executive commentary covering overall performance vs Budget and prior year, the main drivers, and suggested management actions. Then publish the PDF and give me the download link.
```

Why this order works: each prompt asks for one topic, so the agent is less likely to drop a column. By prompt 7 all the data is already in the conversation, so the agent calls `publish_report_pdf` once.

### 7.2 Check the PDF

Click **Download PDF (valid for 1 hour)**. The link is a pre-signed URL on your COS endpoint.

| Check | Expected |
|---|---|
| Title and subtitle | As you typed them |
| One table per section | Executive summary, regions, channels, product, quarterly |
| Regions add up | 20.1 + 14.0 + 19.1 + 22.9 = 76.1 (Actual) |
| Variance columns filled | No `N/A` |
| Summary paragraph above each table | 2 to 4 sentences |
| Executive Commentary section | Headed "Draft - for review", 2 to 3 paragraphs |
| Footer on every page | "Generated ... AI-assisted - Draft - for review" and a page number |
| No `**` characters, no black boxes | Bold text is real bold |

*Screenshot: Preview with the download link*

*Screenshot: Page 1 of the generated PDF*

*Screenshot: Page 2 of the generated PDF with the commentary*

### 7.3 Confirm the file is in COS

This step is optional. The bucket belongs to the instructor, so you may not have console access. If you do: IBM Cloud, then the COS instance, then **Buckets**, then the shared bucket. Open the `reports/` folder. You should see the new PDF. Otherwise, opening the download link is enough to confirm the upload.

*Screenshot: COS bucket reports/ folder with the PDF*

---

## Part 8: Embed the agent in the PA workspace

You need three things: a **key pair**, the **embed values** from wxO, and the **PA integration screen**.

### 8.1 Deploy the agent to the live environment

Embedded chat in production uses the **live** variant.

```
orchestrate agents deploy -n management_report_generator
```

If your CLI asks for an environment, it accepts `--env live` (or `-e live`).

### 8.2 Generate an RSA key pair

Embedded chat is secured with a key pair you create. The **public** key goes to wxO. The **private** key goes to PA.

These commands are the same on Windows (PowerShell) and macOS / Linux. On Windows, `openssl` must be installed (see "Check your tools" in Part 0).

```
openssl genpkey -algorithm RSA -out private_key.pem -pkeyopt rsa_keygen_bits:2048
openssl rsa -pubout -in private_key.pem -out public_key.pem
```

### 8.3 Register the public key in wxO

1. Open the wxO UI, then **Settings**, then the **Embedded chat** (or **Channels**) security section **(confirm label)**.
2. Paste the public key. Copy it to the clipboard with:

**Windows (PowerShell)**

```
Get-Content public_key.pem -Raw | Set-Clipboard
```

**macOS / Linux (Terminal)**

```
cat public_key.pem | pbcopy
```

3. Save and make sure security is **enabled**.

The embedded chat does not start until the keys are configured and validated.

*Screenshot: wxO embedded chat security page with the public key pasted*

### 8.4 Get the embed values

Run:

```
orchestrate channels webchat embed --agent-name=management_report_generator --env live
```

If the command asks for your **CRN**, the automatic lookup failed (this happens when the token has expired or the account cannot read the resource list). Get the CRN from IBM Cloud: **Resource list**, open the Orchestrate instance, and copy the **CRN** from the details panel. Paste it at the prompt.

The command prints a `<script>` block. Copy these four values from it:

| From the script | Needed in PA as |
|---|---|
| `orchestrationID` | `orchestrationID` |
| `hostURL` | `hostURL` |
| `chatOptions.agentId` | `agentId` |
| `chatOptions.agentEnvironmentId` | `agentEnvironmentId` |

You also need your **purchase ID**. It is a value from your PA subscription, not from wxO. **Instructor to confirm** where students find it. The PA documentation calls it `purchaseID` and describes it as "the unique ID for your purchase or subscription".

*Screenshot: Terminal output of the embed command with the four values marked*

### 8.5 Create the configuration file

Create `embed-agent.json` in the `lab-1203` folder (UTF-8, file name must not end in `.txt`):

```
{
  "orchestrationID": "<ORCHESTRATION_ID>",
  "hostURL": "https://ca-tor.watson-orchestrate.cloud.ibm.com",
  "agentId": "<AGENT_ID>",
  "agentEnvironmentId": "<AGENT_ENVIRONMENT_ID>",
  "perspectives": [
    "pa-home",
    "pa-administration",
    "dashboard",
    "modeling",
    "modeling-home"
  ],
  "purchaseID": "<PURCHASE_ID>"
}
```

**Important format note.** The PA documentation shows the values nested inside a `wxo_agent` object, with an `authKey` field. In the lab environment that format was rejected with the error **"Invalid WXO config, orchestrationID is required"**. The format above, with all fields at the top level and **no** `authKey`, is what worked. The private key goes in a separate field (next step).

The `perspectives` list controls where the agent appears. The PA documentation also lists `pa-report` and `pa-plan-contribute`. These two were not tested in the lab.

Check that the file is valid JSON:

**Windows (PowerShell)**

```
python -c "import json;json.load(open('embed-agent.json'));print('valid')"
```

**macOS / Linux (Terminal)**

```
python3 -c "import json;json.load(open('embed-agent.json'));print('valid')"
```

### 8.6 Enter the configuration in PA

1. Sign in to Planning Analytics.
2. In the top bar, open the **Administration** menu (the dropdown next to the gear icon) and choose **Administration**. **(confirm label)**
3. Open **Integrations**.
4. In the list, select **IBM watsonx Orchestrate**. Its status should be **Active**.
5. Select **Custom instance**.
6. Under **Configuration value**, clear any text, then paste the JSON:

**Windows (PowerShell)**

```
Get-Content embed-agent.json -Raw | Set-Clipboard
```

**macOS / Linux (Terminal)**

```
cat embed-agent.json | pbcopy
```

7. Under **Embedded chat authorization key**, paste the **raw** private key, including the `-----BEGIN PRIVATE KEY-----` and `-----END PRIVATE KEY-----` lines and real line breaks. Do **not** convert line breaks to `\n`. Copy it with:

```
cat private_key.pem | pbcopy
```

8. Click **Test**. Fix any error message, then click **Apply**.

After pasting the private key, copy something harmless to overwrite your clipboard.

*Screenshot: PA Administration > Integrations list with IBM watsonx Orchestrate selected*

*Screenshot: Custom instance form with Configuration value and Embedded chat authorization key (blur the key)*

*Screenshot: Successful Test result*

### 8.7 Test inside PA

1. Open a perspective you included, for example **Home**.
2. Open the chat (assistant) panel.
3. Confirm the greeting comes from your agent **(confirm how to tell)**.
4. Run Prompt 1 from Part 4.
5. Run Prompt 7 from Part 7 and open the link.

*Screenshot: Agent chat panel open in a PA perspective*

*Screenshot: PDF link returned inside PA*

### Things to know about the embedded agent

- The agent connects to PA through **one shared account** (the `pa_mcp_credentials` Basic connection). Every user of the embedded agent therefore sees the data that account can see. PA's per-user data security does not apply to the agent. Decide if that is acceptable before using real data.
- Pre-signed links are valid for one hour. After that, ask the agent to produce a new report.
- PA's documentation says OAuth-type connections are not supported in embedded web chat. Basic and key_value connections are what this lab uses.

---

## Part 9: Troubleshooting

| Symptom | Cause | Fix |
|---|---|---|
| `The token found for environment ... is missing or expired` | Login expired | Repeat Part 1.3 (hidden prompt, `activate`, remove the variable) |
| Windows: `orchestrate` or `pip` is not recognized | The virtual environment is not active in this PowerShell window | Run `cd lab-1203` then `.venv\Scripts\Activate.ps1` |
| Windows: `running scripts is disabled on this system` when activating `.venv` | PowerShell execution policy | Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`, then activate again |
| Windows: `python` opens the Microsoft Store or is not found | Python is not installed or not on PATH | Install Python (Part 0, "Check your tools"), then open a new PowerShell window |
| Windows: `openssl` is not recognized | OpenSSL is not installed or not on PATH | Install it (Part 0), reopen PowerShell, or run `$env:Path += ";C:\Program Files\OpenSSL-Win64\bin"` |
| Windows: `&&` is not a valid statement separator | Older Windows PowerShell 5.1 does not support `&&` | Run the commands on separate lines, as the Windows blocks in this guide do |
| `401 Unauthorized` when importing the toolkit | Connection is not Basic, or password is wrong | `orchestrate connections list`. Recreate as `--kind basic` with username `apiKey` |
| `orchestrate chat ask` stops after 90 to 180 seconds | First calls are slow | Test in the wxO preview panel instead |
| Agent says it cannot use Data Explorer | The Revenue cube is not pre-analysed | The agent uses `execute_mdx_and_get_view` instead. Nothing to fix |
| PA PDF tool returns `HTTP 500` from `exploration-analysis-report` | A server-side PA error, reproduced with valid input | Use `publish_report_pdf` (Parts 5 and 6). Report the 500 to IBM if you need that tool |
| `AttributeError: 'dict' object has no attribute 'rows'` | The tool treated tables as objects | Keep the dict-to-`ReportTable` loop in `publish_report_pdf` |
| Variance columns show `N/A` or Budget row missing | Conflicting instructions, or the agent dropped data | Use the variance rule in Part 3.1. Ask for the Budget version explicitly |
| Regions do not add up to Total Company | The agent looked up a region by alias | Use member IDs 100, 200, 300, 400 in the prompt |
| Black box in the PDF text | A character that Helvetica cannot draw | Keep `_safe()` in the tool |
| `**text**` shown in the PDF | Markdown not converted | Keep `_inline()` in the tool |
| "Draft - for review" appears twice | Agent put the label in the commentary | Keep the instruction not to add it, and the filter in the tool |
| `AccessDenied` or signature errors from COS | Wrong keys, endpoint or region | Re-check the five values with your instructor and re-run Part 5.2 |
| `NoSuchBucket` | Wrong `COS_BUCKET` or endpoint | Re-run Part 5.2 with the correct values |

| Symptom | Cause | Fix |
|---|---|---|
| PA shows "Invalid WXO config, orchestrationID is required" | Wrong JSON shape, or leftover text in the box | Use the flat format in Part 8.5. Clear the box with Ctrl+A (Windows) or Cmd+A (macOS), then paste |
| Embedded chat does not load | Public key not registered or security not enabled in wxO, or the private key does not match | Redo Part 8.2 and 8.3 with a fresh pair, and update PA |
| `orchestrate channels webchat embed` asks for a CRN | Automatic lookup returned 401 | Re-activate the environment, or paste the CRN from IBM Cloud |

---

## Part 10: Clean-up

Delete local key files and temporary files:

**Windows (PowerShell)**

```
Remove-Item -Force -ErrorAction SilentlyContinue private_key.pem, public_key.pem, embed-agent.json, tools\test.pdf
```

**macOS / Linux (Terminal)**

```
rm -f private_key.pem public_key.pem embed-agent.json
```

The shared COS bucket and credential belong to the instructor. Do not delete them.

In wxO:

1. Remove the test agent and tools if the account is shared:

```
orchestrate agents remove -n management_report_generator -k native
orchestrate tools remove -n publish_report_pdf
orchestrate toolkits remove -n pa_reporting
orchestrate connections remove -a pa_mcp_credentials
orchestrate connections remove -a cos_reports
```

2. Delete the wxO API key you created in Part 1.2 (Settings, API details).

In PA:

1. Administration, Integrations, IBM watsonx Orchestrate: switch back to **Default instance** and apply.
2. Revoke the PA API key if it was created for this lab.

---

## Appendix A: File layout at the end of the lab

```
lab-1203/
├── .venv/
├── agents/
│   └── management_report_generator.agent.yaml
├── tools/
│   ├── publish_report_pdf.py
│   └── requirements.txt
└── embed-agent.json          (delete in Part 10)
```

## Appendix B: Resources created

| Where | Name | Type |
|---|---|---|
| wxO | `lab-1203` | ADK environment (local name) |
| wxO | `pa_mcp_credentials` | Basic connection (team scope, draft and live) |
| wxO | `pa_reporting` | Remote MCP toolkit, 8 tools |
| wxO | `cos_reports` | key_value connection |
| wxO | `publish_report_pdf` | Python tool |
| wxO | `management_report_generator` | Native agent |
| IBM Cloud | shared COS bucket | Created by the instructor (Appendix D) |
| PA | Integrations, IBM watsonx Orchestrate | Custom instance configuration |

## Appendix C: Command quick reference

| Task | Command |
|---|---|
| Activate environment | `orchestrate env activate lab-1203 --api-key "$WXO_KEY"` |
| List connections | `orchestrate connections list` |
| Add a connection | `orchestrate connections add -a <APP_ID>` |
| Configure a connection | `orchestrate connections configure -a <APP_ID> --env draft --type team --kind <basic or key_value>` |
| Set Basic credentials | `orchestrate connections set-credentials -a <APP_ID> --env draft -u <USER> -p "$PASSWORD"` |
| Set key_value credentials | `orchestrate connections set-credentials -a <APP_ID> --env draft -e "KEY=value"` |
| Add a remote MCP toolkit | `orchestrate toolkits add --kind mcp --name <NAME> --description "<TEXT>" --url <URL> --transport streamable_http --tools "<LIST>" --app-id <APP_ID>` |
| Import a Python tool | `orchestrate tools import -k python -f <FILE> -r <REQUIREMENTS> --app-id <APP_ID>` |
| Import or update an agent | `orchestrate agents import -f <YAML>` |
| Deploy an agent | `orchestrate agents deploy -n <NAME>` |
| Get embed script | `orchestrate channels webchat embed --agent-name=<NAME> --env live` |

## Appendix D: Instructor setup for the shared COS bucket

Do this once before the class. Students never see these steps.

### D.1 Create the bucket

1. In IBM Cloud, open **Resource list**, then **Storage**, then your **Cloud Object Storage** instance.
2. On **Buckets**, click **Create bucket**.
3. Choose **Create a Custom Bucket** (not "Quickly get started").
4. Set:

| Setting | Value |
|---|---|
| Unique bucket name | for example `24retail-mgmt-reports` |
| Resiliency | Regional |
| Location | the same region as wxO (lab: `ca-tor`) |
| Storage class | Smart Tier |
| Public access | **Off** (the agent returns time-limited links instead) |

5. Click **Create bucket**.

*Screenshot: Create bucket page with Custom Bucket card*

*Screenshot: Bucket list showing the new bucket, public access = No*

### D.2 Create a service credential with HMAC keys

The upload tool uses the S3-compatible API, which needs **HMAC** keys.

1. In the COS instance, open **Service credentials** in the left menu.
2. Click **New credential**.
3. Set:

| Setting | Value |
|---|---|
| Name | `lab-1203-cos` |
| Role | **Writer** |
| Service ID | Create new Service ID |
| Advanced options, **Include HMAC Credential** | **On** |

4. If the dialog shows "No instance found, manage credentials with Secrets Manager", the Secrets Manager integration is optional. The exact steps used to get past it were not recorded during the build, so confirm them yourself. If the console will not create the credential, use the CLI command below.
5. Click **Add**.
6. Expand the new credential. Find the `cos_hmac_keys` section. You need `access_key_id` and `secret_access_key`.

If the console will not create it, the CLI can (this needs `ibmcloud login` first):

**Windows (PowerShell)**

```
'{"HMAC":true}' | Set-Content hmac.json -Encoding ascii
ibmcloud resource service-key-create lab-1203-cos Writer --instance-name "<COS_INSTANCE_NAME>"
--parameters "@hmac.json"
Remove-Item hmac.json
```

**macOS / Linux**

```
ibmcloud resource service-key-create lab-1203-cos Writer --instance-name "<COS_INSTANCE_NAME>"
--parameters '{"HMAC":true}'
```

*Screenshot: New credential dialog with "Include HMAC Credential" switched on (Advanced options)*

*Screenshot: Expanded credential showing cos_hmac_keys (blur both values)*

### D.3 Find the endpoint

Use the **public regional** endpoint for your bucket's region. For `ca-tor`:

```
https://s3.ca-tor.cloud-object-storage.appdomain.cloud
```

For another region, open **Endpoints** in the COS left menu and copy the **Public** endpoint of your bucket's region.

*Screenshot: COS Endpoints page with the public regional endpoint marked*

### D.4 Hand out the values

Give students the five values from Part 5.1: endpoint, region, bucket, access key id and secret. Use a handout or slide, not the guide. Create a new credential for each class and delete it afterwards.

### D.5 Shared-bucket considerations

- Report file names are `reports/<timestamp>-<random id>.pdf`, so students do not overwrite each other.
- Anyone with the keys can list, read and (with the Writer role) delete every student's files. This is acceptable for the sample 24Retail data. Do not use real data.
- Download links expire after 1 hour. The bucket has public access off.
- After the class: delete the credential, then empty and delete the bucket (or add a lifecycle rule to expire `reports/`).